

proxy_cache_challenge.sh

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Authority:** Helix Constitution §11.4.27 (Challenges), §11.4.69 (sink-side positive evidence), §11.4.107 (forgeable-header discipline), §11.4.3 (honest SKIP), §11.4.68 (no fail-open)

Overview

Anti-bluff Challenge that proves the **live HTTP proxy** (default `http://localhost:53128`) caches. A cacheable plain-HTTP URL is fetched **twice** through the proxy; the **authoritative** proof is a Squid `TCP_*HIT` result code for that URL in the Squid `access.log`, asserted via `assert_cache_hit` from `tests/lib/evidence.sh`. An X-Cache header or an Age field alone is **forgeable** and is never accepted as the verdict — it is captured only as supplementary evidence. If the `access.log` is not reachable/readable, the challenge SKIPS with the closed-set reason `topology_unsupported` (§11.4.3); it never fakes a PASS.

Prerequisites

- The base proxy stack UP with the HTTP proxy on 53128.
- `curl`, `bash`, `awk`, `grep`; `tests/lib/evidence.sh` present (sourced).
- The authoritative branch additionally requires the Squid `access.log` to be **readable** by the running user. In the reference rootless-container topology the log is owned by a container subuid (mode `0640`) and is **not** readable by the host user — the challenge then honestly SKIPS.
- READ-ONLY: plain proxy client + a read of the `access.log` — never touches any container.

Usage

```
bash challenges/scripts/proxy_cache_challenge.sh
# override the cache url / log path / evidence dir:
CACHE_URL=http://example.com/ \
SQUID_ACCESS_LOG=/path/to/access.log \
CHALLENGE_EVIDENCE_DIR=/path/to/evidence \
  bash challenges/scripts/proxy_cache_challenge.sh
```

Environment: HTTP_PROXY_URL (default http://localhost:53128), CACHE_URL (default http://example.com/), SQUID_ACCESS_LOG (override the resolved path), LOG_DIR (default ./logs; the compose file maps container /var/log/squid here), CHALLENGE_EVIDENCE_DIR (default qa-results/challenges/<run-ts>), CURL_MAX_TIME (default 20).

Access-log resolution

SQUID_ACCESS_LOG → \$LOG_DIR/access.log (from env) → .env LOG_DIR → ./logs/access.log (the docker-compose.yml default mount).

Exit codes

Code	Meaning
0	PASS — a Squid TCP_*HIT for the URL was found in a readable access.log.
1	FAIL — proxy could not fetch the reachable URL, or a readable log showed no TCP_*HIT for a URL fetched twice.
3	SKIP — outage (URL unreachable) or the access.log is not readable (topology_unsupported, §11.4.3).

Anti-bluff notes

- The connectivity precondition uses proxy_conn_verdict so a site outage SKIPS while a proxy-broken-but-site-reachable-directly case FAILs.
- Via: / Age: headers are logged as **supplementary** only; the verdict rests on the data-plane TCP_*HIT result code, per §11.4.107.
- Unreadable log ⇒ honest topology_unsupported SKIP with the `ls -ln` ownership captured as the reason — never a fabricated cache PASS.

Outputs

<evidence-dir>/cache/cache_evidence.txt (double-fetch codes, Via/Age headers, connectivity verdict, and either the `assert_cache_hit` verdict or the `ls -ln` unreadable-log evidence) plus the two raw header dumps.

Related scripts

- `proxy_forward_http_challenge.sh`, `proxy_socks5_challenge.sh` — forwarding proofs.
- `run_proxy_challenges.sh` — the bank runner.
- `tests/lib/evidence.sh` — `assert_cache_hit` and the sourced helpers.

Last verified: 2026-07-01 (live run: precondition PASS, authoritative SKIP topology_unsupported — access.log owned by container subuid 100012, mode 0640, not readable by uid 1000).